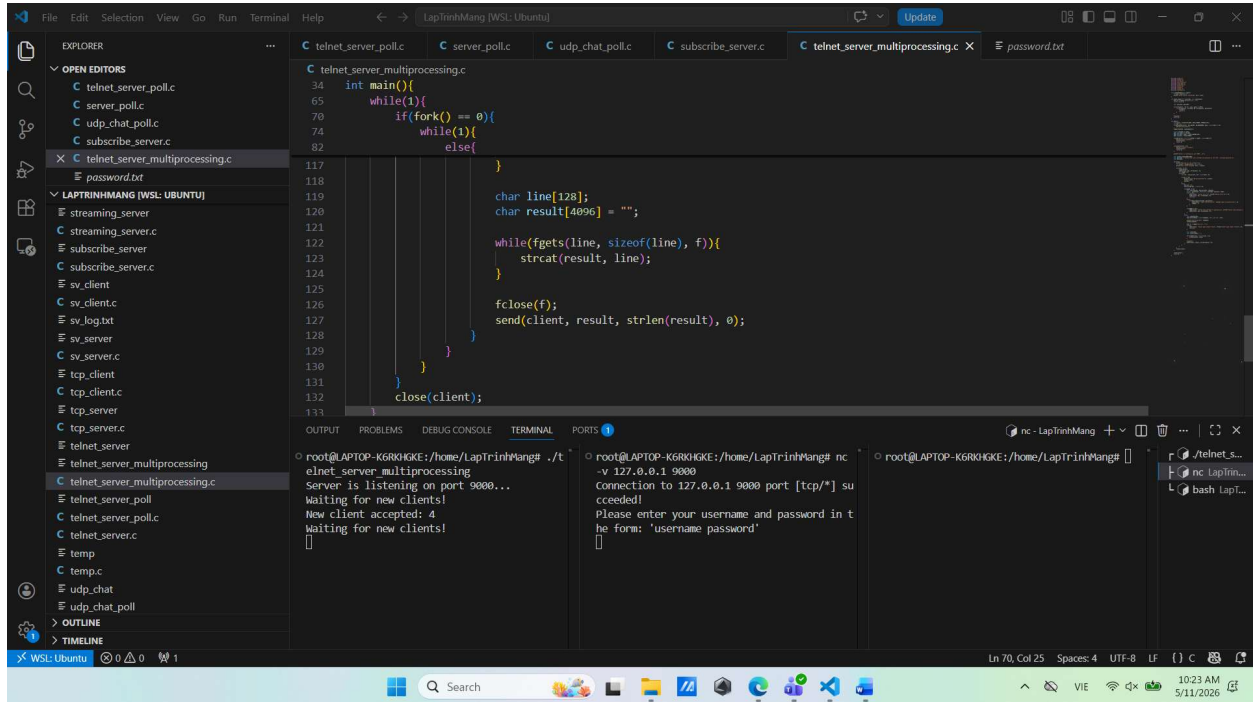


Báo cáo BTVN 5_5

Bài 1:

- Khi có client kết nối đến, server sẽ hỏi username và password của client:



The screenshot shows a Visual Studio Code editor window with a C program named `telnet_server_multiprocessing.c` open. The code is a multi-threaded server that listens on port 9000. When a client connects, it forks a child process to handle the connection. The child process reads a line of input from the client and sends it back. The parent process then closes the client connection and continues to listen for new clients.

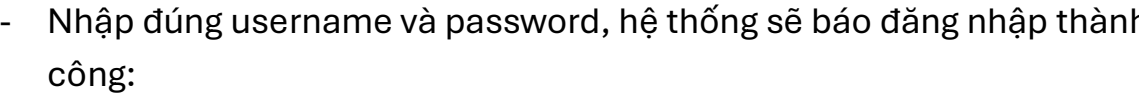
```
34 int main(){
65     while(1){
70         if(fork() == 0){
74             while(1){
82                 else{
117             }
118         }
119         char line[128];
120         char result[4096] = "";
121         while(fgets(line, sizeof(line), f)){
122             strcat(result, line);
123         }
124         fclose(f);
125         send(client, result, strlen(result), 0);
126     }
127 }
128 }
129 }
130 }
131 }
132 }
133 }
134 }
135 }
136 }
137 }
138 }
139 }
140 }
141 }
142 }
143 }
144 }
145 }
146 }
147 }
148 }
149 }
150 }
```

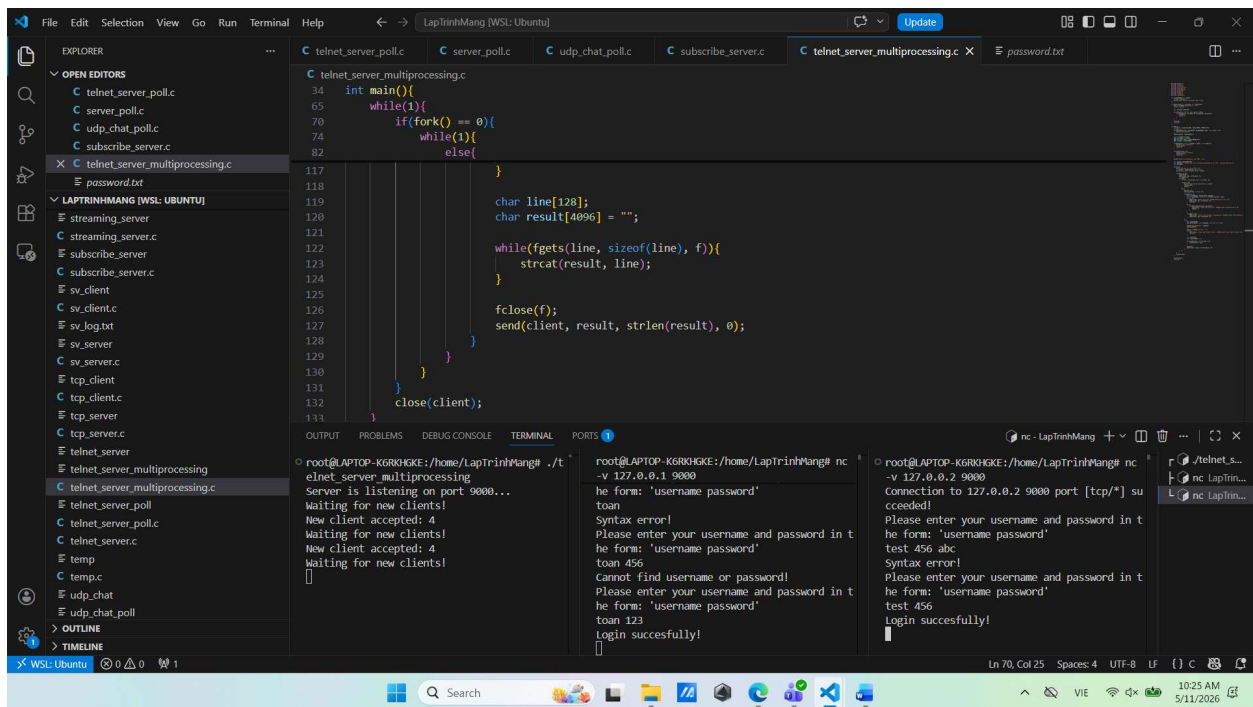
The terminal output shows the server running and listening on port 9000. It receives a connection from 127.0.0.1 and prompts the user to enter a username and password. The user enters 'username' and the server responds with 'password'.

```
root@LAPTOP-K6RKHGKE:/home/LapTrinhMang# ./telnet_server_multiprocessing
Server is listening on port 9000...
Waiting for new clients!
New client accepted: 4
Waiting for new clients!
[ ]
```

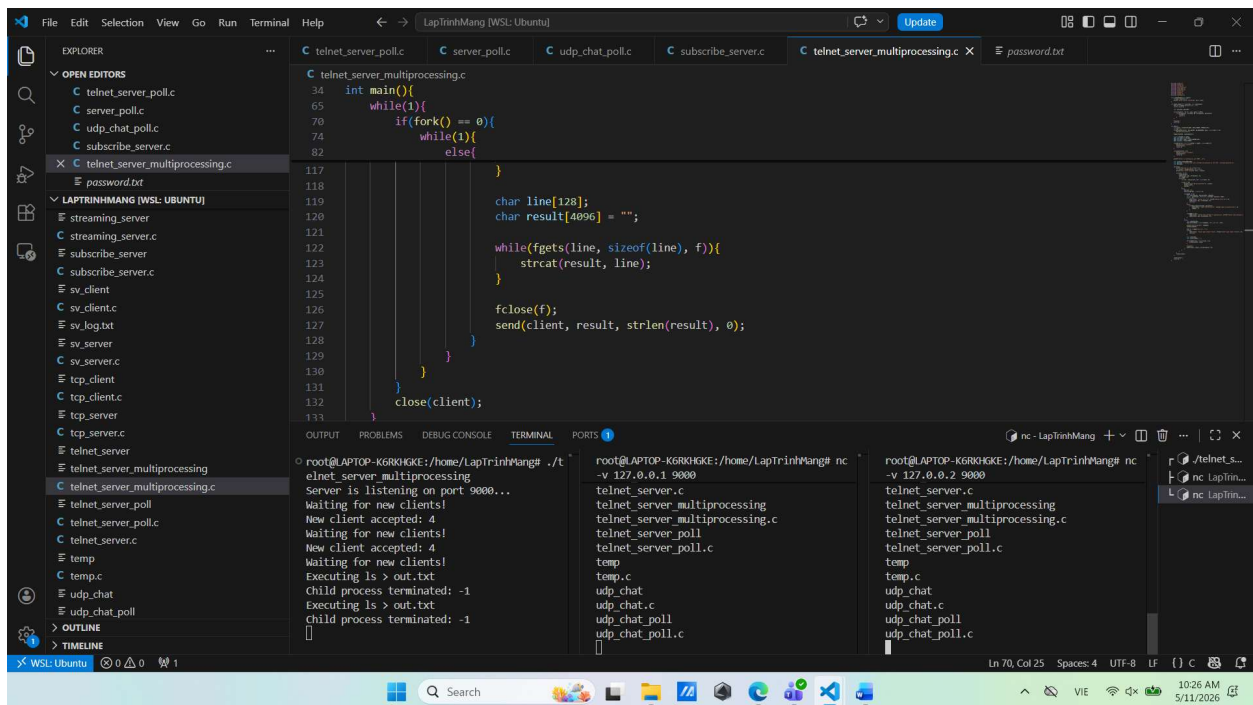
```
nc -LapTrinhMang -v 127.0.0.1 9000
Connection to 127.0.0.1 9000 port [tcp/*] succeeded!
Please enter your username and password in the form: 'username password'
[ ]
```

- Nếu nhập thừa hoặc thiếu, server sẽ báo syntax error:

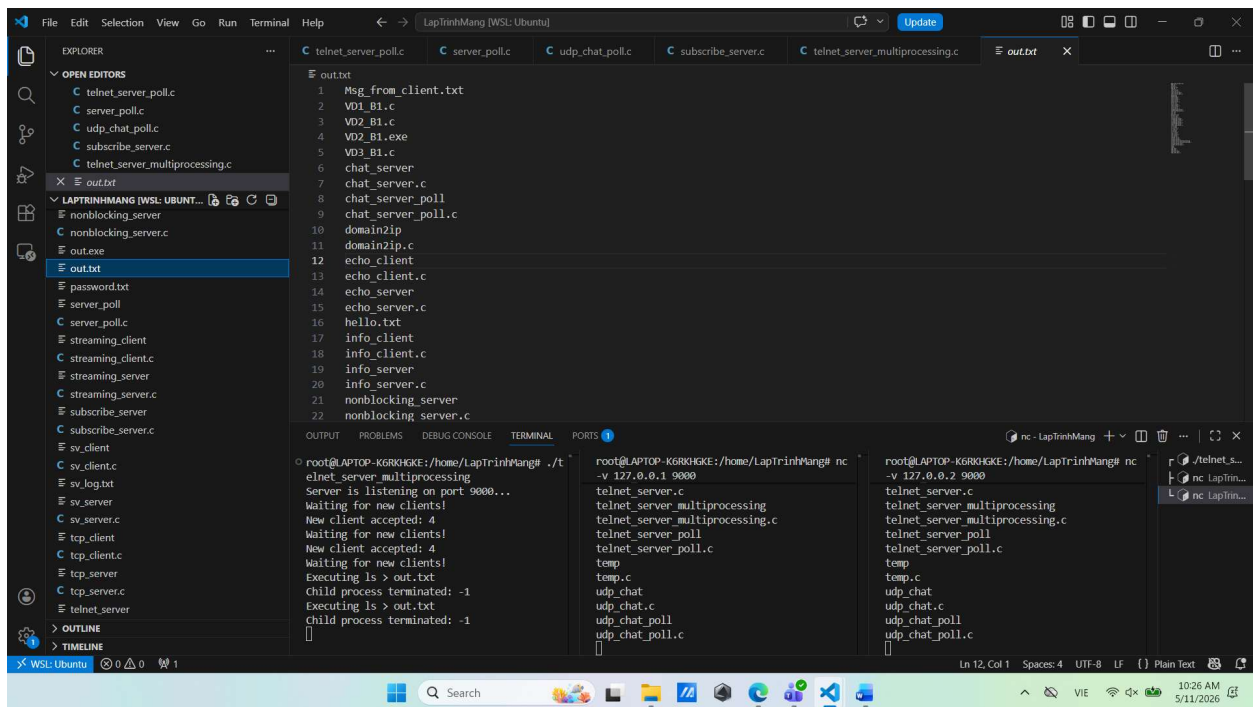




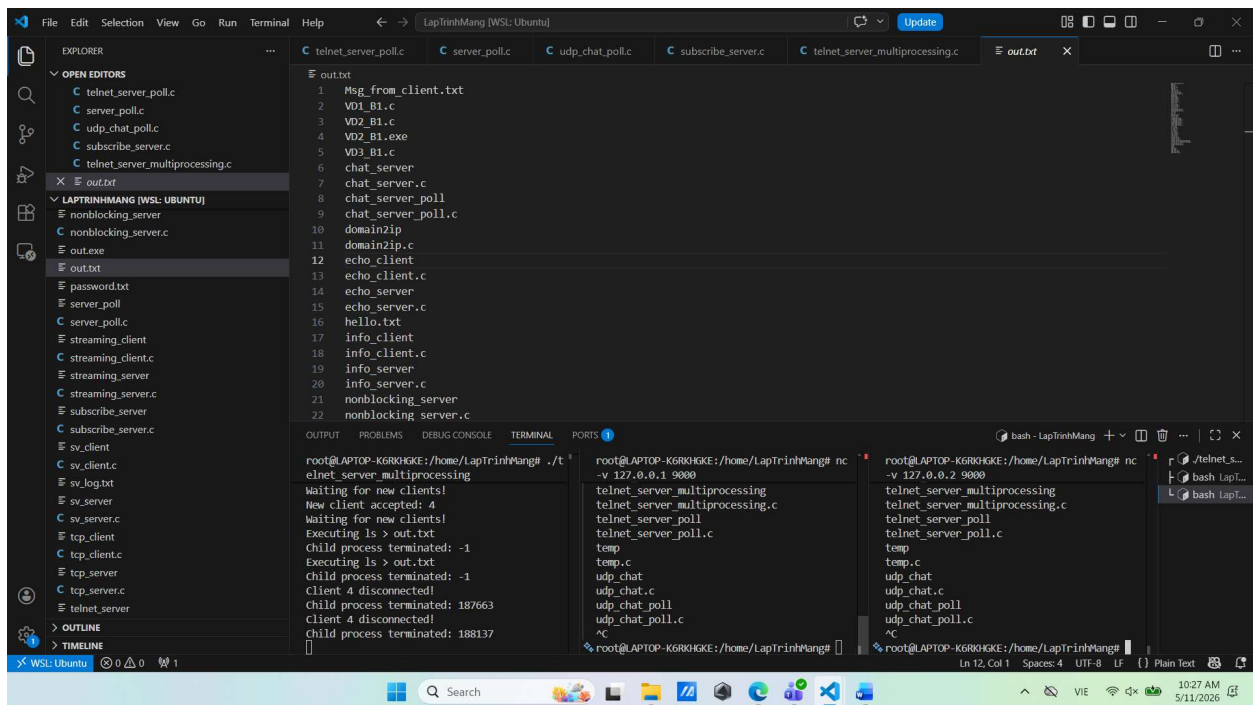
- Client gửi lệnh, server thực hiện lệnh và trả kết quả về cho client:



- Kết quả chạy cũng được lưu vào file out.txt:

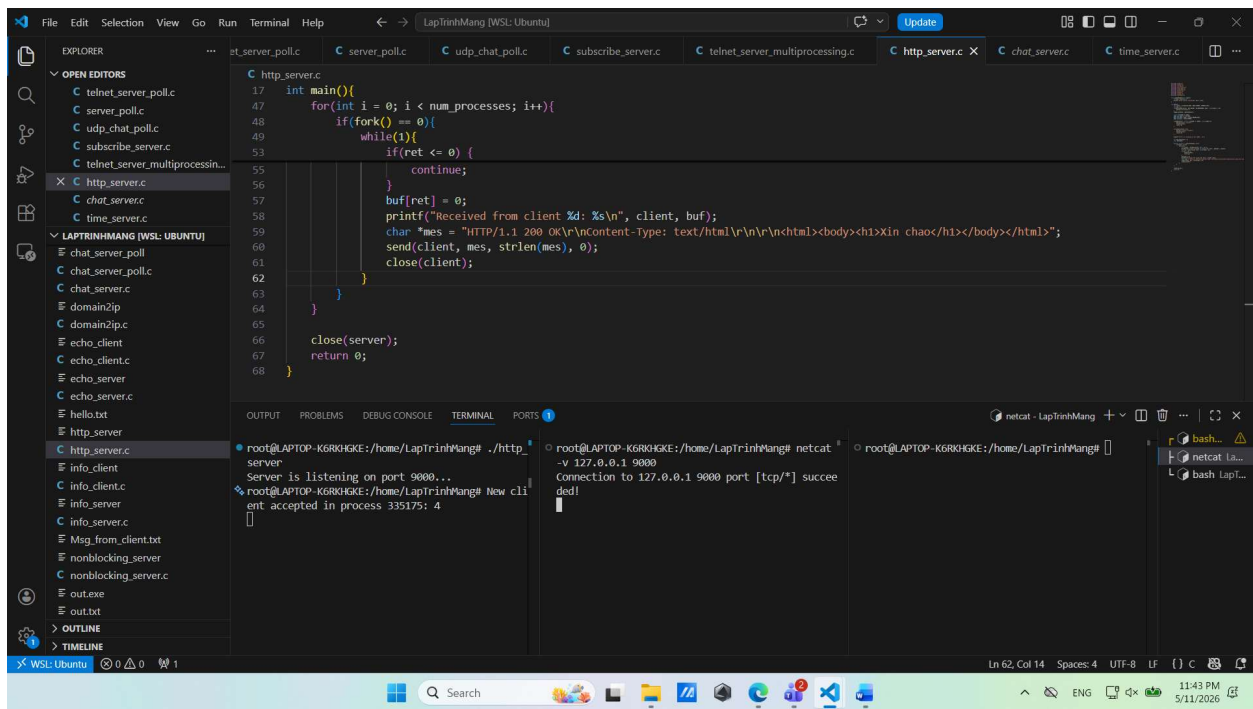


- Server thông báo khi client disconnect:

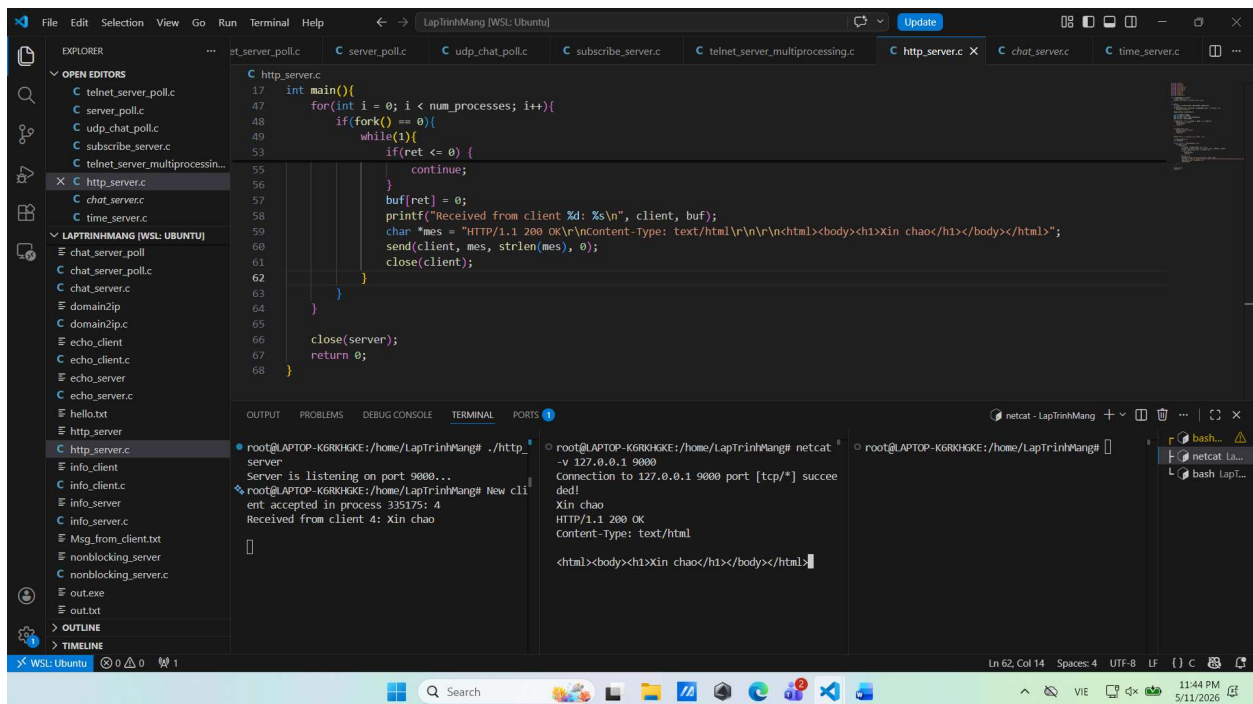


Bài 2:

- Chạy file `http_server` và dùng netcat để tạo client kết nối đến server ở cổng 9000:

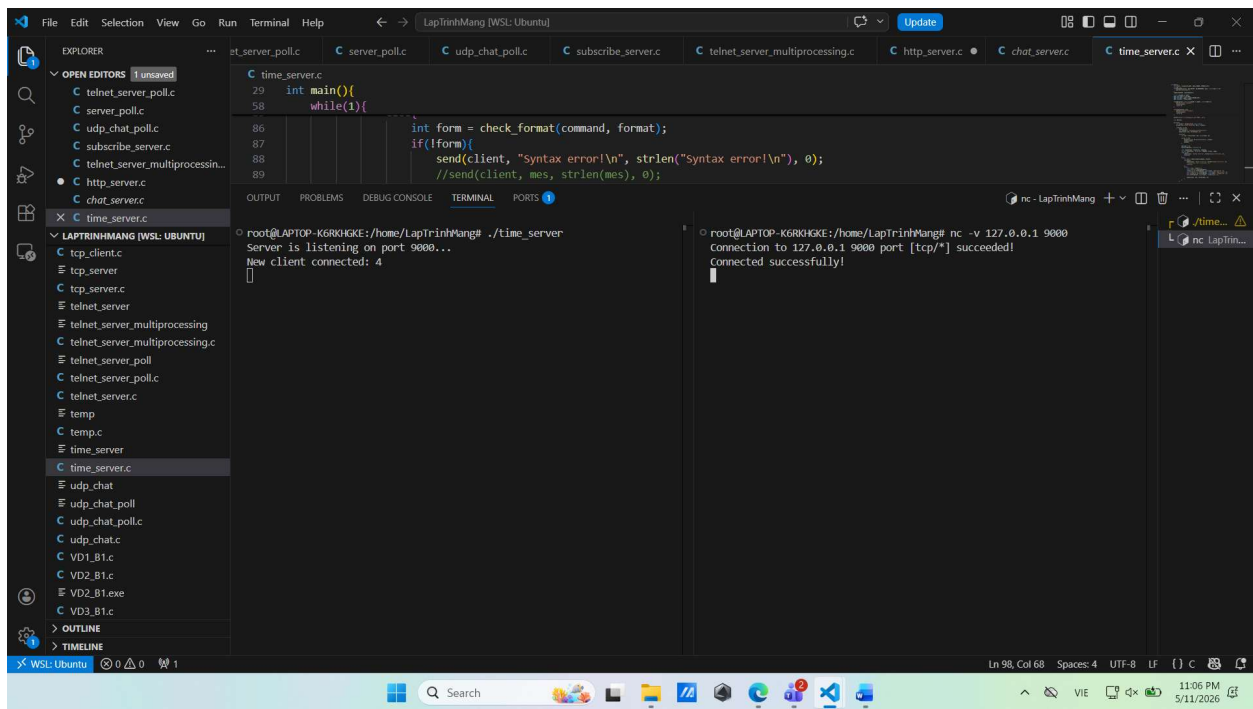


- Khi client gửi tin nhắn server hiển thị ra màn hình và gửi phản hồi tới client:

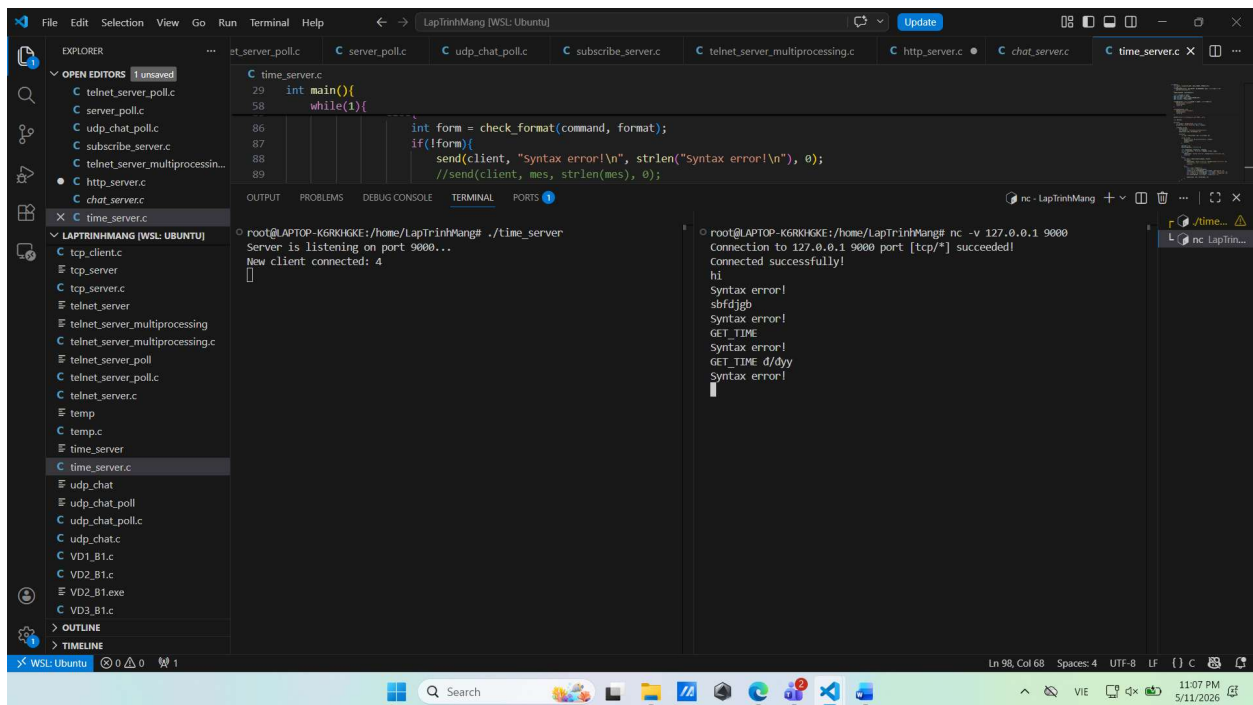


Bài 3:

- Chạy file time_server và dùng netcat tạo client kết nối đến cổng 9000:



- Nhập sai cú pháp sẽ báo syntax error:



- Nhập đúng yêu cầu sẽ trả về thời gian theo đúng định dạng đã yêu cầu:

The screenshot shows the Visual Studio Code interface with a C program named `time_server.c` open in the editor. The code is as follows:

```
1 int main()
2 {
3     while(1){
4         int form = check_format(command, format);
5         if(!form){
6             send(client, "Syntax error!\n", strlen("Syntax error!\n"), 0);
7             //send(client, mes, strlen(mes), 0);
8         }
9     }
10 }
```

The terminal output shows the server listening on port 9000 and receiving a connection from 127.0.0.1. The client sends various commands, and the server responds with "Syntax error!" for invalid commands and the current time for valid ones.

```
root@LAPTOP-K6RKHGKE:/home/LapTrinhMang# ./time_server
Server is listening on port 9000...
New client connected: 4

root@LAPTOP-K6RKHGKE:/home/LapTrinhMang# nc -v 127.0.0.1 9000
Connection to 127.0.0.1 9000 port [tcp/*] succeeded!
Connected successfully!
hi
Syntax error!
sbfjdjgb
Syntax error!
GET_TIME
Syntax error!
GET_TIME d/dy
Syntax error!
GET_TIME dd/mm/yyyy
11/05/2026
GET_TIME dd/mm/yy
11/05/26
GET_TIME mm/dd/yyyy
05/11/2026
GET_TIME mm/dd/yy
05/11/26
```

- Chấp nhận kết nối từ nhiều client:

The screenshot shows the Visual Studio Code interface with the same C program `time_server.c` open. The terminal output shows the server listening on port 9000 and receiving multiple connections from different clients. The client sends various commands, and the server responds with "Syntax error!" for invalid commands and the current time for valid ones.

```
root@LAPTOP-K6RKHGKE:/home/LapTrinhMang# ./time_server
Server is listening on port 9000...
New client connected: 4

root@LAPTOP-K6RKHGKE:/home/LapTrinhMang# nc -v 127.0.0.1 9000
Connection to 127.0.0.1 9000 port [tcp/*] succeeded!
Connected successfully!
hi
Syntax error!
sbfjdjgb
Syntax error!
GET_TIME
Syntax error!
GET_TIME d/dy
Syntax error!
GET_TIME dd/mm/yyyy
11/05/2026
GET_TIME dd/mm/yy
11/05/26
GET_TIME mm/dd/yyyy
05/11/2026
GET_TIME mm/dd/yy
05/11/26

root@LAPTOP-K6RKHGKE:/home/LapTrinhMang# netcat -v 127.0.0.2 9000
Connection to 127.0.0.2 9000 port [tcp/*] succeeded!
Connected successfully!
GET_TIME dgkn
Syntax error!
GET_TIME dd/mm/yyyy
11/05/2026
```

- Server trả về thông báo khi client ngắt kết nối:

